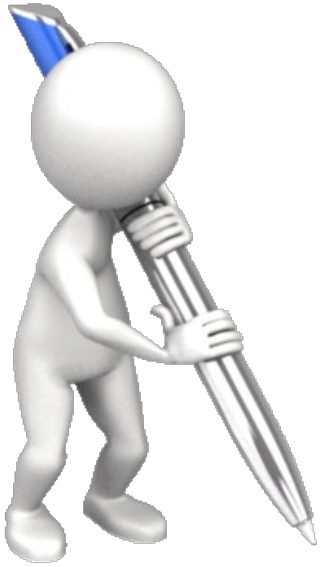
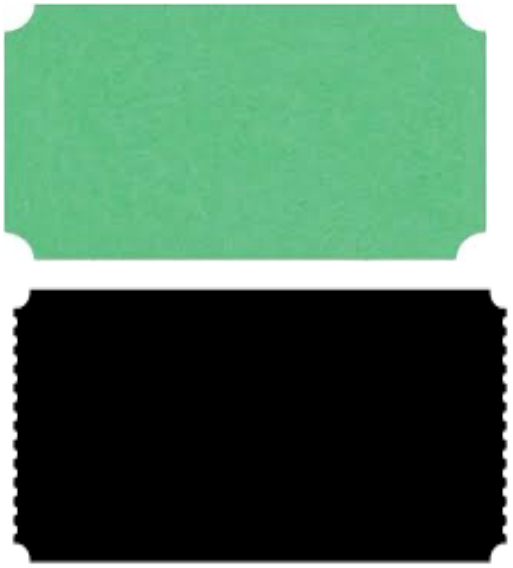
# Data Science in Action: A Clinical Trial



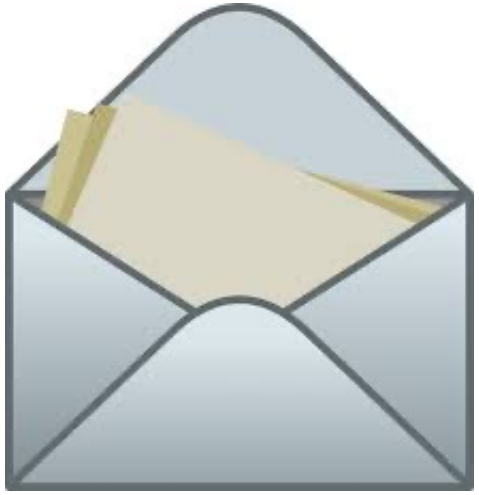
# #1 Give Consent

- Print and sign your name agreeing to be a subject in this clinical trial
- **Giving consent** means agreeing to be a subject in a randomized study and you are aware that you will be asked to read something and then asked to make a food choice. **There are no right or wrong answers, and you can stop at anytime.**



# #2 Be Randomized

- Reach into the bag and pull out a paper ticket at **Random**
  - If Green you are in Green Group
  - If Black you are in Black Group



# #3 Intervention

- Read item in Envelope that corresponds to your Group

Green Intervention

# Why is fruit important?

Fruit provides vitamins, minerals, dietary fibre and many phytonutrients (nutrients naturally present in plants), which help your body stay healthy. Since fruit is low in calories (energy) relative to many other foods, including it every day can help prevent excessive weight gain. Eating fruit may also help protect against chronic diseases such as heart disease, stroke and some types of cancers. Different fruit can help protect the body in different ways, so it's important to choose a variety of colours everyday such as:

- green (such as apples, grapes)
- orange (such as oranges, mango, peaches)
- yellow and red (such as strawberries, banana).
- purple (blueberries, plums, grapes)

## How much fruit does a young person need each day?

| 2-3 years | 4-8 years | 9-18 years |
|---|---|---|
| 1 | 1 ½ | 2 |

**Whole fruit is a healthy option when you want something sweet**

Black Intervention

# The Candy Man

Who can take a sunrise (who can take a sunrise)
Sprinkle it with dew (sprinkle it with dew)
Cover it with choc'late and a miracle or two
The Candy Man (the Candy Man)
Oh, the Candy Man can (the Candy Man can)
The Candy Man can
'Cause he mixes it with love
And makes the world taste good (makes the world taste good)

Who can take a rainbow (who can take a rainbow)
Wrap it in a sigh (wrap it in a sigh)
Soak it in the sun and make a groovy lemon pie
The Candy Man (the Candy Man)
The Candy Man can (the Candy Man can)
The Candy Man can
'Cause he mixes it with love
And makes the world taste good (makes the world taste good)

# #4 Choose

- Take candy or fruit

NOTE: you can only take one or the other, not both





# #5 Record your choice

- Put a tick mark on the board under your group whether you chose **candy** or **fruit**











# Kids in each intervention group recorded their choice throughout the day



# What do you think the Research Question was in this Clinical Trial?

A. Do people like candy?

B. Can people read advertisements?

C. If people read an advertisement about healthy eating are they more likely to choose fruit?

Answer: C

# Does the intervention work?

- Do you think there is enough evidence based on the **current data** to say that people who read about healthy eating are more likely to choose fruit?
- How could we use our data to answer this question?

# Results at the End of the Day

(throughout the day, the numbers fluxuated as additional kids were enrolled)




N=63 kids participated

39 randomized to GREEN
27 randomized to BLACK

- 12/39 = 30.7% of the Green group chose fruit after the intervention
- 7/27 = 25.9% of the Black group chose fruit after the intervention

Difference: 4.8% higher proportion of fruit chosen in Green group but
95% Confidence interval (-20.3%, 30.0%)
p-value = 0.88

**No statistically significant evidence that intervention worked**

```
IN R:
prop.test(x = c(12, 7), n = c(39, 27))
2-sample test for equality of proportions with continuity correction
data:  c(12, 7) out of c(39, 27)
X-squared = 0.022741, df = 1, p-value = 0.8801


alternative hypothesis:
two.sided95 percent confidence interval:
-0.2026907  0.2995568

sample estimates:
prop 1    prop 2
0.3076923 0.2592593
```